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Project Title:	Quenching of Particle-Gas Combustible Mixtures Using Electric Particulate Suspension (EPS) and Dispersion Methods		
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Program/Discipline:	COMBUSTION SCIENCE		
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	This NASA research was carried out with the goal of developing a new technology in powder combustion that supports NASA’s space science effort in microgravity and fire safety, referred to here as EPS (Electric Particulate Suspension). The accurate measurement of flame-quenching distance in dusts is an important parameter in fire safety as well as in advancing laminar flame theory as it is closely related to the characteristic flame thickness and structure. Achieving a uniform dust concentration has also been identified as a key problem to be resolved in dust flame studies and a major technical obstacle in experimental work on dust quenching. A 5-tiered EPS combustion experiment meeting these performance requirements has been rigorously tested in microgravity providing up to 5 ignition data points per test in the 2.2 second drop tower facility at NGRC (NASA Glenn Research Center). The new electrostatic approach is fundamentally simple utilizing multiple (parallel plate) test cells driven by a single high voltage (kV)-low current DC		

Task Description:	source and spark ignition. Each EPS cell generates a steady-state dust cloud requiring less than one gram of powder for a quenching test. Complete quenching distance curves can be generated versus fuel: oxidant ratio. The primary objective of this study is to validate the EPS method for quenching and combustion testing in microgravity by evaluating improvements in EPS performance and expanding ranges of testing. Specific objectives include extending the test envelope of powder combustion to richer mixtures, decreasing cloud stratification, increasing flame-quenching test distances, and the measurement of burning velocity in dust clouds. Performance objectives of EPS in microgravity include generating higher particle concentrations and reduced stratification effects compared to normal gravity. Another objective is to identify the minimum electric field strength correlations necessary to maintain suspensions in both 0-g and 1-g. The above goal and objectives were met.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	ISU's Electric Particulate Suspension (EPS) provides a unique electrostatic technique for dispersing and controlling particle suspensions (clouds) in vacuum and pressurized environments both in normal and zero gravity fields. The EPS combustion method has potential as a 'benchmark' design establishing a fire safety standard for measuring both flame propagation in dusty gas mixtures (in gravity and microgravity) and inert powder quenching distance of combustible gases. A compact "suitcase" EPS design is also possible as portable fire safety instrument. The fundamental research will be of particular interest to those working in laboratories and industry across the United States, particularly those concerned with high technology combustion research and also safe handling of powders. The EPS method also opens the way for new powder handling technologies that include applications involving reaction chemistry and particle transport processing. EPS supports steady-state (flowing) and batch (confined) processing of powders. Potential applications in which EPS has been demonstrated include chemistry and physical reactions of powders (5 μm to several mm), powder separation processing, enhanced solids heat transfer, and pneumatic transport of powders below the saltation (minimum transport) velocity.
	A new experiment referred to as EPS (Electric Particulate Suspension) designed for quenching distance measurements of combustible powder suspensions has been taken from concept to working prototype and validated in microgravity studies over a six year period. A special 5-tiered EPS rig was extensively tested in 42 microgravity (<10-4 g) drops in the 2.2 second drop tower (79 ft 1 in) at NASA Glenn Research Center (NGRC). The 5-tiered experiment is designed for quick disassembly and easy cleaning/loading of powder samples after a single end plate is removed. Once assembled, the 5-tiered rig with 10 inch diameter stainless plates fit snugly inside the combustion chamber located on the "Pig-1" A-Frame. The closed system test cells are preloaded with different powder/oxidant ratios providing 5 quenching data points in a single drop that determine quenching distance. With this method complete curves of ignition-quenching distance for powders (e.g. aluminum) and oxidants (e.g. air) can be obtained. Three independent methods are used to measure particle number (mass) concentration with EPS: 1) Measure the weight (mass) of power suspended in cell of known volume 2) Measure laser attenuation through the suspension applied to the Beer-Lambert law to calculate the particle number density of the suspension 3) Calibrate the maximum mass suspended using 1) against current flow in external EPS circuit – used only for laboratory studies with normal gravity The computer-controlled burn sequence during the drop permits the electrostatic-driven suspension to adjust to microgravity conditions over the first 1.6 to 1.8 seconds of free fall followed by the spark ignition. The energy delivered to the 5 sparks is independently controlled from the EPS high voltage supply (0 to 25 kV driving the suspension) by a second high voltage DC source (0 to 25 kV) connected to capacitance banks. Particle concentration profiles are recorded for a single (center) test cell before and during the drop using a laser-photodiode scanner controlled by the onboard DDACS (Droppable Data Acquisition System) computer. An onboard CCD video camera (30 frames/sec) provides graphic visualizations of cloud formations and intense sparks (~10 J) accompanied by (often) brilliant powder burns. The onboard photodiode array performed to specifications giving rapid and multiple scans of cloud profiles in a few ms during the drop. However, the arrays could not produce the high special resolution provided by the single laser ground-based scanner. Results from tests on aluminum, glass, and copper powder show that microgravity improves suspension quality and expands the combustion test range of EPS to significantly higher powder concentrations over normal gravity. Decreased cloud stratification is also observed compared to normal gravity. A summary of the quenching results is as follows: • Flammability and quenching curves can be generated with EPS in normal gravity over a wide range of lean to rich mixtures; however, achieving rich concentrations is limited compared to microgravity • High energy sparks appear to be necessary for ignition of very rich mixtures of aluminum-air mixtures tested in microgravity. • Quenching distances measured in microgravity using the new EPS 5-tiered rig are consistent with (EPS) values measured in normal gravity – i.e. validating the multiple test cell method. • The EPS method can be applied to either cylindrical or linear devices for measuring flame propagation in admixtures of combustible gases and dusts for quenching and non-quenching studies. The methods can be adapted to microgravity. General Observations of Powders in Microgravity: Tests in microgravity with EPS confirm that higher powder concentrations can be produced compared to normal gravity for the same electric field strengths. Stratification in suspensions (asymmetry of concentration profiles from gravity) is eliminated at 0-g. A criterion to minimize stratification is developed based on laser profile scans of suspensions in microgravity. Onboard videos show that powders initially adhering to surfaces will not "float off" in microgravity but are held to walls and electrode surfaces by naturally occurring and charged induced cohesion/adhesion forces, i.e. "floating" from walls depends on particle initial force conditions. As predicted, microgravity also shows a significant reduction in

Task Progress:	the minimum electric field strength necessary to drive suspensions with an increase by factors of 2-4 in the maximum particle-cloud concentration over normal gravity. Newfound phenomena observed in microgravity includes: (1) The appearance of “cellular” flow structures in high concentration suspensions at high electric field strength-to-gravity ratios indicating a cloud dynamic instability; (2) The growth of “stringer and spider web” structures in post-combustion processes for aluminum-air mixtures at high fuel/air ratios. Similar but larger column structures have been artificially grown in our laboratory using high voltage fields and fine glass particles. The concept of “excess electric field strength” as a parameter is introduced to correlate maximum-of-mass concentration data in any gravity field. The excess field is the additional electric field strength to overcome gravity and induced electrical attraction. Advantages of EPS Method: • Simple parallel plate geometry and knob voltage control: Particle concentration, particle speed (gas-solids mixing), particle-particle-wall collision rate can be controlled • Generation of uniform steady-state powder suspensions over wide range of particle sizes (5-1000 μm). Cloud uniformity is achieved through particle-collisions. • Electric suspensions are readily generated in both microgravity and gravity environments • Electric suspensions can be used in both open ended (ducted) flows and closed (batch) type systems as well as inparticle diffusion-control systems utilizing particle- particle and particle-wall collisions • Low power consumption (mW/cc) utilizes the combination of high voltage (kV)-low current (micro-amp) DC supply • Compact “suitcase” size experiments are suitable for land rovers and are portable as astronaut packs • Electric suspensions are readily produced in powders having finite (semi-insulating) surface/bulk electrical conductivity e.g. glass, sand, magnesium, aluminum, copper, iron • Electric suspensions can be generated in vacuum and high pressure gases and insulating liquids (eg mineral oil) • High temperature EPS is possible if below gas ionization and electrode shorting. Our spark-ignition experiments are transient with collapse of the suspension occurring following the spark and passage of the flame. Suspension collapse (with combustion) may or may not be an issue in microgravity depending on the application as steady-state or transient. Suspensions otherwise run indefinitely in steady-state. Restrictions on EPS Method: • Generally EPS is well suited for small scale and portable applications (rover and suitcase experiments) as well as for extended length ducted flow experiments because of practical limitations set by electrode separations of a few centimeters (e.g. 5-6 cm). Staging is possible for greater powder throughputs. • Fines below 5 microns and cohesive powders (e.g. 40 micron pulverized coal) can be difficult to suspend by EPS – and as well by most any pneumatic method. Use of acoustic breakup of agglomerates has been successful in producing good suspensions for powders such as pulverized coal. • Highly insulating powders may not suspend, e.g Teflon coated glass. A small but finite surface or bulk electrical conductivity is required. • Use of a very high voltage (> 30 kV) sources may prove to be a restriction in some applications from corona discharge or sparking. Off-the-shelf battery driven programmable high voltage power supplies (fits in hand) are available for EPS using typical inputs of 30 VDC (~1 A) – for example, the drop tower study utilized 28 V DC batteries. • Operation of EPS in glass enclosures are subject to humidity > 30-40 % because of increased surface electrical conductivity (e.g. summer months and high humidity areas….Winter months and low humidity <20 % works best). In addition to identifying a new method for generating curves of quenching distance of powders for fire safety, the EPS method is equally suitable for testing non-reacting powders as quenching agents in admixtures of combustible gases. Early studies on quenching by inert particles using EPS have been published previously. McGill-ISU Collaboration: The original NASA proposal included a collaboration effort between McGill University and ISU team in quenching research. A shared goal has been to expand the understanding of the fundamentals of powder combustion (burning velocity and quenching) using very different methods (in microgravity). In contrast to the EPS method, the McGill approach is based on a dust dispersion system utilizing a combustion tube assembly, optical dust concentration monitoring system, and filtering and venting system. The ISU team designed and built two rigs for McGill that could be used to provide calibration curves of particle concentration using laser attenuation. Unfortunately, a technical problem in dispersing the very small size of McGill aluminum powder (mean diameter ~ 4 μm) using EPS together with a costly equipment failure of a vibration table proved to be problematic. This together with the long development period of ISU’s EPS experiment (5+ years) did not allow for an exchange of experimental results as proposed.
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